

Data Identification, Sourcing Strategy & Initial Analysis Planning

1. Introduction

Building on Week 1, where the problem was defined as *predicting customer purchase trends in skincare products*, this week focuses on identifying relevant data sources and planning for initial exploratory analysis. The aim is to ensure that the chosen data aligns with the Beauty & Wellness theme and supports the hypotheses.

2. Data Sourcing Strategy

- **Approach:**
 - Review publicly available datasets, APIs, and industry reports.
 - Prioritize sources that provide customer demographics, purchase behavior, and product trends.
 - Ensure data accessibility, reliability, and frequency of updates.
- **Criteria for Selection:**
 - Relevance to skincare/beauty products.
 - Availability of demographic and transactional variables.
 - Data quality (completeness, accuracy, consistency).
 - Ease of access (open datasets, APIs, CSV downloads).

3. Potential Data Sources

- **Government Databases:**
 - Ministry of Health & Family Welfare reports (India).
 - Statista datasets on skincare market growth.
- **Public APIs:**
 - Kaggle datasets (e.g., skincare product reviews, e-commerce sales).
 - Open Beauty & Wellness APIs (product catalogs, reviews).
- **Social Media Data:**
 - Twitter API for sentiment analysis on skincare brands.
 - Instagram trends (hashtags like #skincare, #organicbeauty).
- **E-commerce Platforms:**
 - Amazon/Flipkart public datasets (customer reviews, ratings).
 - Meesho product category insights.

4. Initial Exploratory Analysis Plan

- **Descriptive Statistics:**
 - Mean, median, mode of purchase frequency.
 - Distribution of spending amounts.
- **Visualization Techniques:**
 - Histograms (age vs product preference).

- Line charts (seasonal demand trends).
- Word clouds (social media reviews).
- **Key Variables:**
 - Age, gender, location.
 - Product category, purchase frequency.
 - Seasonal factors, sentiment scores.

5. Data Quality & Preprocessing

- **Quality Checks:**
 - Missing values, duplicates, inconsistent formats.
 - Outlier detection in spending amounts.
- **Preprocessing Steps:**
 - Standardize categorical variables (e.g., product categories).
 - Normalize numerical data (spending, frequency).
 - Text cleaning for social media reviews (stopwords, emojis).

6. Alignment with Week 1 Hypotheses

- Age/gender variables → test influence on product preference.
- Seasonal data → test impact on demand fluctuations.
- Sentiment analysis → validate hypothesis about organic product preference among younger customers.

7. Conclusion

This sourcing and analysis plan ensures that the data collected will be relevant, reliable, and aligned with the Week 1 problem statement. The structured approach provides a foundation for deeper analysis in the coming weeks.